

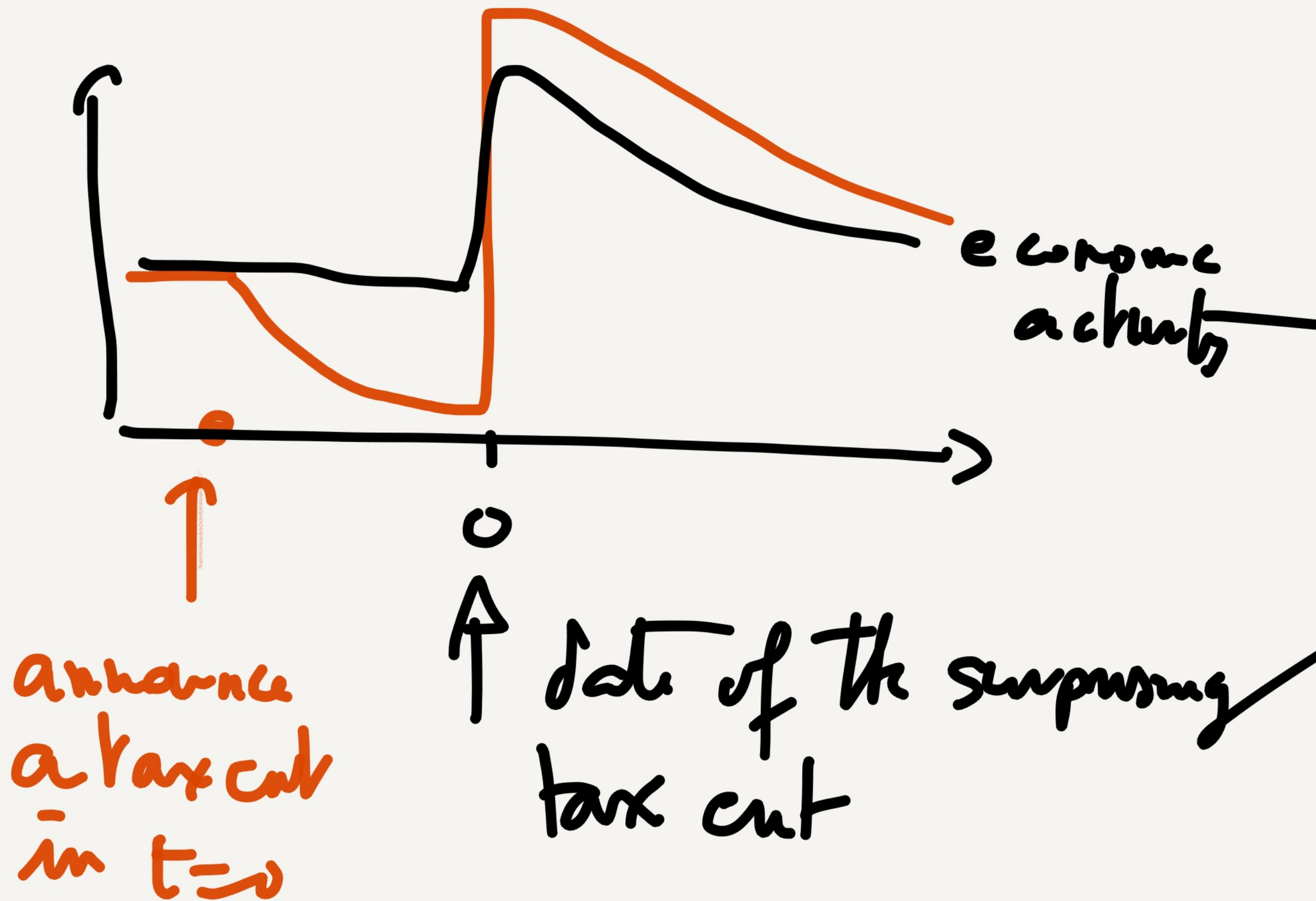
FRANK PORTER
2016-2017

M1 MACRO

Group A

Session 16

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.



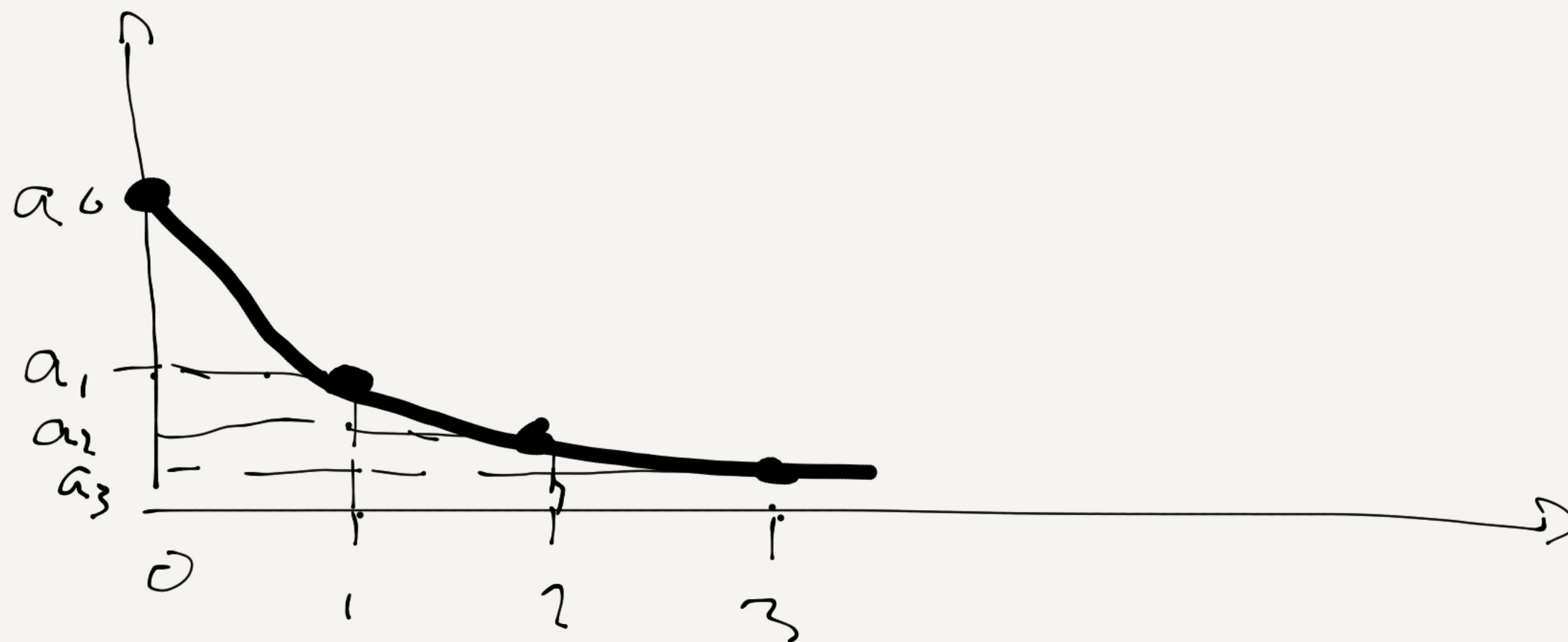
IMPULSE RESPONSE FUNCTION

$$\hat{Y}_t = \sum_{i=0}^{\infty} a_i \varepsilon_{t-i} ; \quad \left| \frac{\partial \hat{Y}_{t+h}}{\partial \varepsilon_t} = a_h \right|$$

↓
dynamic multiplier

$$\hat{Y}_{t+h} = \sum_{i=0}^{\infty} a_i \varepsilon_{t+h-i}$$

$$\hat{L}_t = a_0 \varepsilon_t + a_1 \varepsilon_{t-1} + a_2 \varepsilon_{t-2} + \dots$$



Do the shock: $\varepsilon_0 = 1$ ($\varepsilon_{t < 0} = 0$, $\varepsilon_{t > 0} = 0$)
 $\hat{L}_0 = a_0$, $\hat{L}_1 = a_1$, $\hat{L}_2 = a_2 \dots$

A nice result

$$|p| < 1$$

$$y \quad \boxed{X_t = p X_{t-1} + \varepsilon_t}$$

$$= p (p X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = p^2 \overset{X_{t-2}}{\varepsilon_{t-1}} + \varepsilon_t$$

$$= \dots$$

$$= \varepsilon_t + p \varepsilon_{t-1} + p^2 \varepsilon_{t-2} + p^3 \varepsilon_{t-3} \dots$$

$$\rightarrow X_t = pB X_t + \varepsilon_t$$

$$(1 - pB) X_t = \varepsilon_t$$

$$X_t = \frac{\varepsilon_t}{1 - pB}$$

$$\hookrightarrow \varepsilon_t + pB \varepsilon_t + p^2 B^2 \varepsilon_t + p^3 B^3 \varepsilon_t \dots$$
$$= (1 + pB + (pB)^2 + (pB)^3 \dots) \varepsilon_t$$

$$\hookrightarrow \boxed{\frac{1}{1 - pB} = 1 + pB + (pB)^2 + (pB)^3 \dots}$$

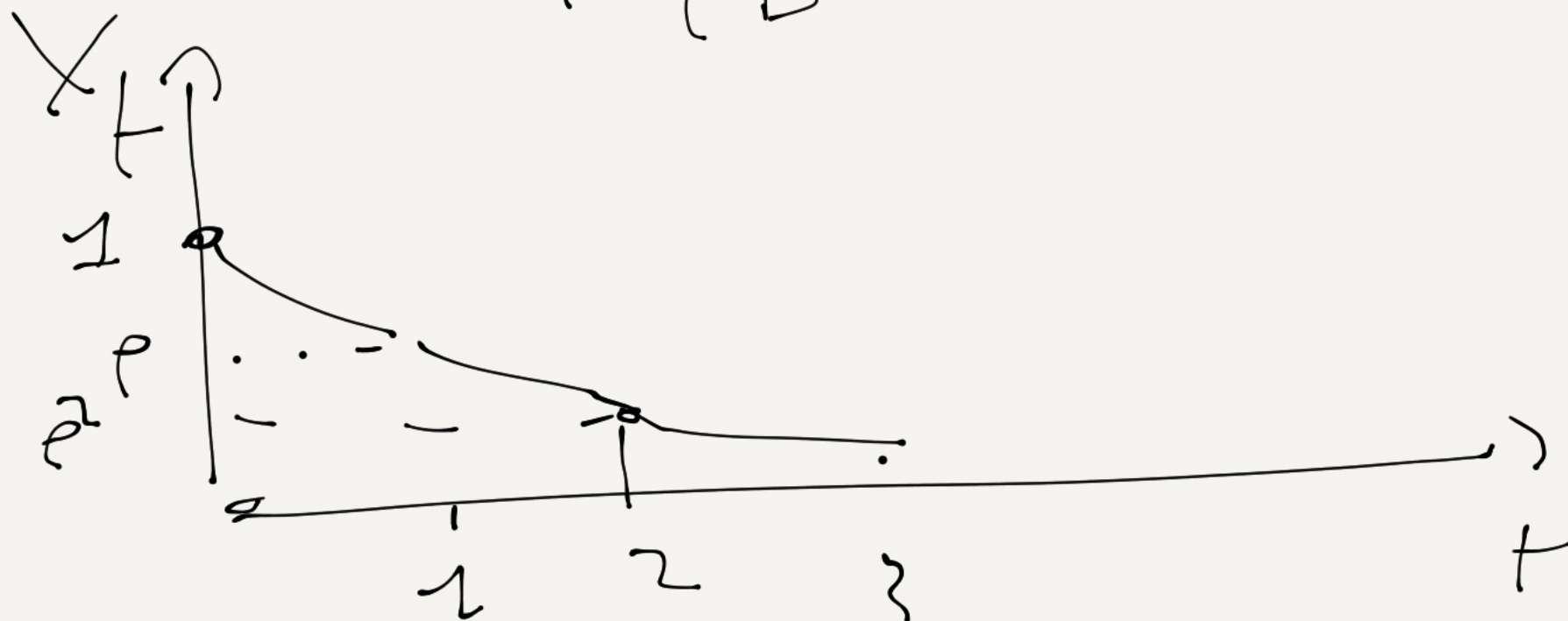
$$X_t = \rho X_{t-1} + \varepsilon_t$$

$$|\rho| < 1$$

$$X_t = \rho \beta X_t + \varepsilon_t$$

$$(1 - \rho \beta) X_t = \varepsilon_t$$

$$X_t = \frac{\varepsilon_t}{1 - \rho \beta} = \varepsilon_t + \rho \varepsilon_{t-1} + \rho^2 \varepsilon_{t-2} + \rho^3 \varepsilon_{t-3} + \dots$$



$$\varepsilon_0 = 1$$

$$\varepsilon_t > 0 \Rightarrow$$

$$\varepsilon_t < 0 \Rightarrow$$

In the Adg. Cost model

$$\hat{L}_t = \lambda_1 \hat{L}_{t-1} + \mu \tilde{w}_t \quad (1)$$

$$\tilde{w}_t = \rho \tilde{w}_{t-1} + \varepsilon_{wt} = \mu \frac{\varepsilon_{wt}}{1 - \rho\beta}$$

$$(1) : \hat{L}_t (1 - \lambda_1 \beta) = \frac{\varepsilon_{wt}}{1 - \rho\beta}$$

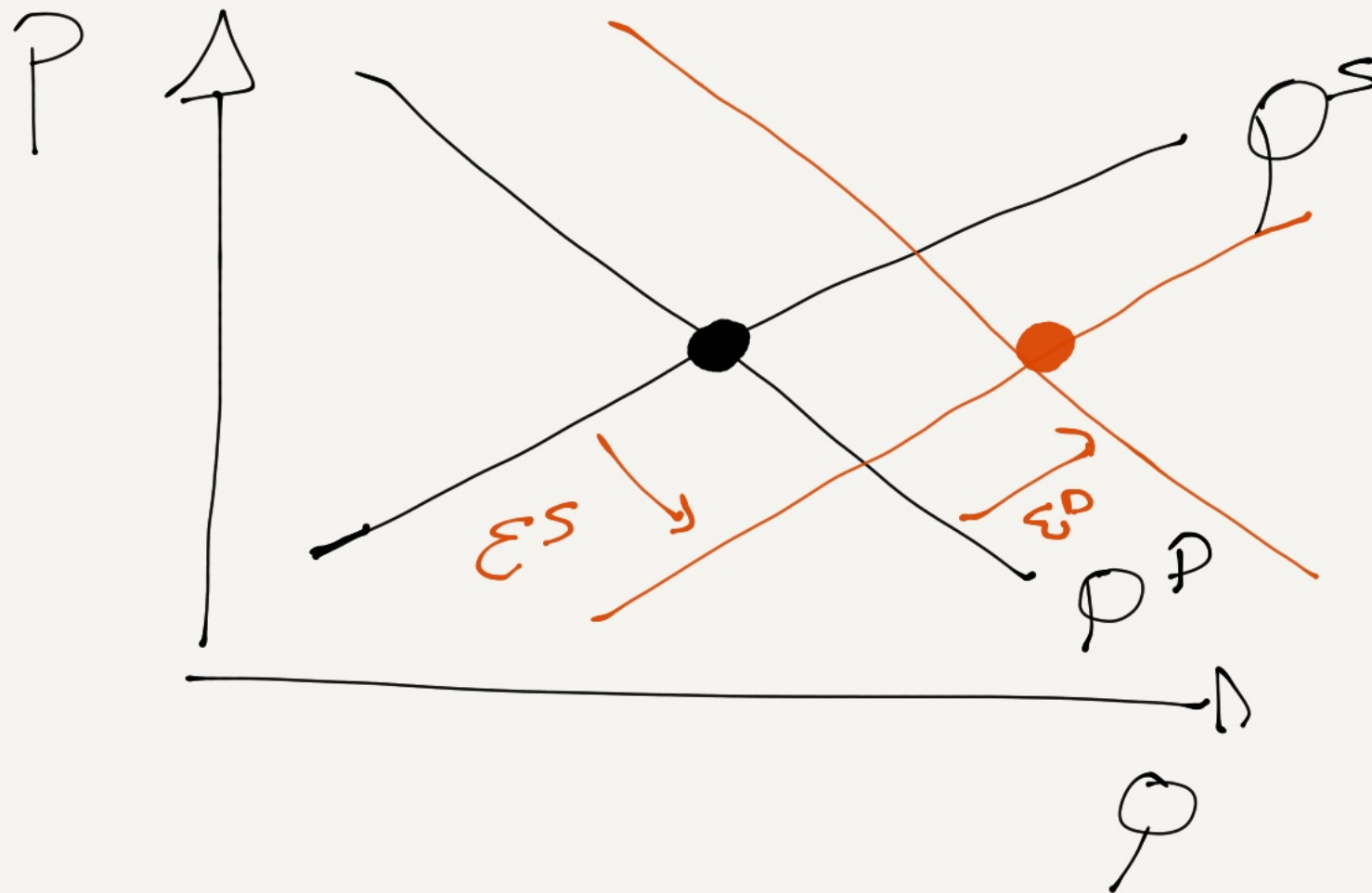
$$\hat{L}_t = \frac{\mu \varepsilon_{wt}}{(1 - \rho\beta)(1 - \lambda_1 \beta)} = \left[\frac{A}{1 - \rho\beta} + \frac{\beta}{1 - \lambda_1 \beta} \right] \mu \varepsilon_{wt}$$

IDENTIFICATION

$$\begin{cases} Q_t^S = \beta P_t + \varepsilon_t^S \\ P_t^D = -\alpha P_t + \varepsilon_t^D \end{cases}$$

$$Q_t = a P_t + \varepsilon_t$$

• Data (Q_t, P_t)

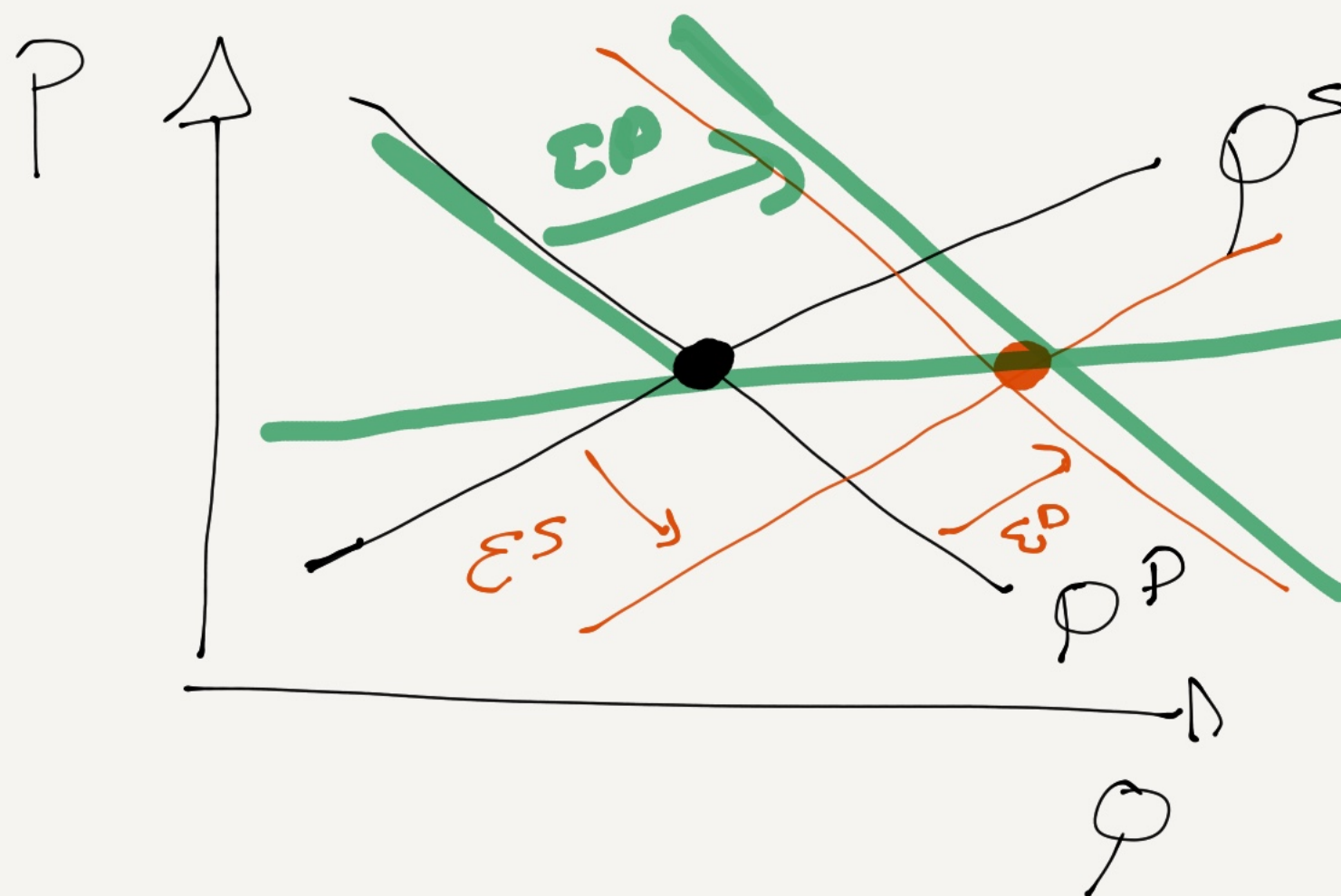


← if I know α, β

IDENTIFICATION

$$\begin{cases} Q_t^S = \beta P_t + \varepsilon_t^S \\ P_t = -\alpha P_t + \varepsilon_t^D \end{cases}$$

• Data (P_t, Q_t)



Q^S
if I know
 α, β

How to estimate α and β ?

$$P_t^S = \alpha P_t + \varepsilon_t^S \quad \longrightarrow \quad \text{OLS} : \varphi \text{ on } P \quad \longrightarrow \quad \alpha?$$

$$P_t^D = -\beta P_t + \varepsilon_t^D \quad \longrightarrow \quad \text{OLS} \quad \longrightarrow \quad \beta?$$

If I know that ε^D is "almost 0"

